

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

# Chapter 5: Quadratic Equations

Topic-wise short notes with solved examples

A quadratic equation has the variable squared. This chapter covers solving it by factorisation and by the formula, giving answers to a required accuracy, and finding the nature of its roots.

## 1 Standard form and roots

- A **quadratic equation** in  $x$  has the form

$$ax^2 + bx + c = 0, \quad a \neq 0$$

- $a$ ,  $b$  and  $c$  are real numbers. If  $a = 0$  the equation is no longer quadratic.
- A value of  $x$  that satisfies the equation is a **root** (solution). A quadratic equation has two roots, which may be equal.
- To check whether a number is a root, substitute it and see whether both sides become equal.

**Example 1.1** If  $x = 2$  is a root of  $3x^2 - px - 2 = 0$ , find  $p$ .

**SOLUTION**

Substitute  $x = 2$ :  $3(4) - 2p - 2 = 0$

$$10 - 2p = 0$$

$$p = 5$$

## 2 Solving by factorisation

- Write the equation in the form  $ax^2 + bx + c = 0$ , clearing fractions and brackets first.
- Split  $bx$  into two terms whose sum is  $b$  and whose product is  $ac$ , factorise, and put each factor equal to zero (**zero product rule**: if  $pq = 0$  then  $p = 0$  or  $q = 0$ ).

**Example 2.1** Solve  $2x^2 - 7x + 6 = 0$ .

**SOLUTION**

$ac = 12$ ; the numbers  $-4$  and  $-3$  have sum  $-7$  and product  $12$ .

$$2x^2 - 4x - 3x + 6 = 0, \text{ so } 2x(x - 2) - 3(x - 2) = 0 \text{ and } (2x - 3)(x - 2) = 0$$

$$x = \frac{3}{2} \text{ or } x = 2$$

**Example 2.2** Solve  $x + \frac{1}{x} = 2\frac{1}{2}$ ,  $x \neq 0$ .

**SOLUTION**

$x + \frac{1}{x} = \frac{5}{2}$ . Multiply by  $2x$ :  $2x^2 + 2 = 5x$ , so  $2x^2 - 5x + 2 = 0$

$$(2x - 1)(x - 2) = 0$$

$$x = \frac{1}{2} \text{ or } x = 2$$

**Example 2.3** Solve  $\frac{x}{x-1} + \frac{x-1}{x} = 2\frac{1}{2}$ .

**SOLUTION**

Let  $\frac{x}{x-1} = y$ . Then  $y + \frac{1}{y} = \frac{5}{2}$ , which gives  $y = 2$  or  $y = \frac{1}{2}$  (as in the previous example).

$$\frac{x}{x-1} = 2 \text{ gives } x = 2x - 2, \text{ so } x = 2$$

$$\frac{x}{x-1} = \frac{1}{2} \text{ gives } 2x = x - 1, \text{ so } x = -1$$

$$x = 2 \text{ or } x = -1$$

### 3 The quadratic formula

- The roots of  $ax^2 + bx + c = 0$  are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- Use the formula when the equation does not factorise, and always when the question says "correct to ... decimal places" or "significant figures".

**Example 3.1** Solve  $x^2 - 7x + 3 = 0$ , giving your answer correct to two decimal places.

**SOLUTION**

$$a = 1, b = -7, c = 3; b^2 - 4ac = 49 - 12 = 37 \text{ and } \sqrt{37} \approx 6.083$$

$$x = \frac{7 \pm 6.083}{2}, \text{ so } x = \frac{13.083}{2} = 6.5415 \text{ or } x = \frac{0.917}{2} = 0.4585$$

$$x = 6.54 \text{ or } x = 0.46$$

#### REMEMBER

For an answer correct to 2 decimal places, work to 3 places and round off only at the end.  
For significant figures, count from the first non-zero digit: 0.4585 correct to 2 significant figures is 0.46.

### 4 Nature of the roots

- The **discriminant** is  $D = b^2 - 4ac$ .

| Discriminant | Nature of the roots                    |
|--------------|----------------------------------------|
| $D > 0$      | Real and unequal                       |
| $D = 0$      | Real and equal, each $= -\frac{b}{2a}$ |
| $D < 0$      | Not real (imaginary)                   |

- If  $D$  is a perfect square (and  $a, b, c$  are rational), the roots are rational; otherwise they are irrational.
- The roots are **real** when  $D \geq 0$ .

**Example 4.1** Without solving, find the nature of the roots of (i)  $3x^2 - 5x + 2 = 0$  (ii)  $x^2 + 4x + 5 = 0$ .

**SOLUTION**

(i)  $D = 25 - 24 = 1 > 0$

(ii)  $D = 16 - 20 = -4 < 0$

**(i) Real and unequal (also rational, since 1 is a perfect square). (ii) Not real.**

**Example 4.2** Find the values of  $m$  for which  $x^2 - 2mx + 9 = 0$  has equal roots.

**SOLUTION**

For equal roots,  $D = 0$ :  $(-2m)^2 - 4(1)(9) = 0$

$4m^2 = 36$ , so  $m^2 = 9$

$m = 3$  or  $m = -3$

## 5 Common mistakes

- Dividing both sides by  $x$  and losing the root  $x = 0$ .
- Reading  $b$  or  $c$  without its sign.
- Dividing only  $\sqrt{D}$  by  $2a$  instead of the whole numerator.
- Rounding too early, which makes the final decimal places wrong.
- Not checking that a solution keeps every denominator non-zero.